Linkmodel: powerlogit

Parametrization

This is the link that map $p \in (0,1)$ into $x \in \Re$, where

$$F^{\beta}(x) = p, \qquad \beta > 0$$

and $F(x)$ is the cummulative distribution function for the logit,

$$F(x) = \frac{1}{1+\exp(-x)}.$$

This link is renormalized so its corresponding density have zero mean and unit variance for every value of β.

Hyperparameters

The parameter β represent the power

$$\beta = \exp(\theta_1)$$

and the prior is defined on θ_1.

The intercept is represented by a quantile level α, where

$$\alpha = \frac{\exp(\theta_2)}{1+\exp(\theta_2)}$$

and the prior is defined on θ_2.

Specification

Use `model="powerlogit"` within `control.link`.

Hyperparameter spesification and default values

doc `Power logit link`

hyper

theta1

hyperid `49131`
name `power`
short.name `power`
initial `0.00123456789`
fixed `FALSE`
prior `normal`
param `0 10`
to.theta `function(x) log(x)`
from.theta `function(x) exp(x)`

theta2

hyperid `49132`
name `intercept`
short.name `p0`

initial `0`
fixed `FALSE`
prior `logitbeta`
param `1 1`
to.theta `function(x) log(x / (1 - x))`
from.theta `function(x) exp(x) / (1 + exp(x))`

pdf `powerlogit`

Example

```
map <- function(x) log(x/(1-x))
imap <- function(x) 1/(1+exp(-x))

power.link <- function(power) {

    qfunc <- function(q) -log(q^(-1/power) -1)
    dfunc <- function(x) power * exp(-x) / (1+exp(-x))^(power + 1)
    pfunc <- function(x) 1/(1+exp(-x))^power

    dq <- 0.005
    qq <- unique(sort(c(0+exp(seq(-10, 0, by = dq)),
                        1-exp(seq(-10, 0, by = dq)))))
    qq <- qq[(qq < 1) & (qq > 0)]
    xx <- qfunc(qq)
    ## total probability, need to account for this in cdf/icdf so it will extrapolate no need
    ## do this in the m & s calcuation
    p <- qq[length(qq)]
    dens <- dfunc(xx)
    dx <- (c(0, diff(xx)) + c(diff(xx), 0))/2
    dens <- (dens * dx) / sum(dens * dx)
    m = sum(xx*dens)
    s = sqrt(sum(xx^2*dens) - m^2)
    xx <- (xx-m)/s
    csum <- p * cumsum(dens)
    cdf <- splinefun(xx, map(csum), method = "monoH.FC")
    icdf <- splinefun(map(csum), xx, method = "monoH.FC")
    return(list(cdf = cdf, icdf = icdf, mean = m, sd = s))
}

n <- 500
size <- 25
power <- 0.25
link <- power.link(power)
x <- rnorm(n, sd = 0.5)
x <- x-mean(x)
intercept <- 1
prob <- imap(link$cdf(intercept + x))
y <- rbinom(n, size = size, prob = prob)

r <- inla(y ~ 1 + x,
```

```
        data = data.frame(y, x),
        family = "binomial",
        Ntrials = size,
        inla.mode = "experimental",
        verbose = TRUE,
        control.inla = list(int.strategy = "eb"),## cmin = 0),
        control.predictor = list(compute = TRUE, link = 1),
        control.fixed = list(prec = 1,
                             remove.names = "(Intercept)"),
        control.family = list(
            ## start with the true values to demonstrate how its done
            control.link = list(
                model = "powerlogit",
                hyper = list(
                    intercept = list(initial = link$cdf(intercept),
                                     prior = "normal",
                                     param = c(0,1)),
                    power = list(initial = log(power),
                                 param = c(0, 1),
                                 fixed = FALSE)))),
        num.threads = "4:1")

summary(r)

## this should be, about, the values of the intercept
print(round(dig = 4, c(intercept = intercept,
                       approximate.intercept.est =
                           as.numeric(link$icdf(r$mode$theta[2])))))
```

Notes

- This link is EXPERIMENTAL, and new default priors will be added in the near future
- Setting the initial value for the hyperparameter "intercept" to infinity, will remove the intercept from the link-model.